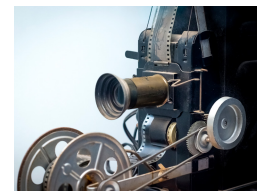


Feature



Geologists on the silver screen—the sequel

As geologists, our interest is piqued when geological processes—with varying degrees of accuracy—are portrayed in movies and especially when a geologist unexpectedly appears on screen, even in a minor role. More than a decade ago, during a coffee break, we started talking about movies that feature geologists. We counted a dozen different movies and began to wonder: how is our profession portrayed in the popular mass media?

Convinced that there could not be that many movies with geologists, we set out to create a list. Starting with movies seen by us or our acquaintances, we soon widened the data collection to online movie databases, including transcribed movie dialogues. We limited our query to British and American movies that have been shown in cinema.

We constrained the definition of a ‘geologist movie’ so that at least one clearly identified geologist—dead or alive—must appear on screen. An example of a movie that does not meet this criterion is *Tarzan and the Jungle Boy* (1968), in which Tarzan helps search for the lost son of a deceased geologist, who does not appear in the movie.

We also chose to differentiate between geologists and closely related geoscientists, such as geophysicists and paleontologists, mainly because these are often identified separately and portrayed as performing different tasks. For example, the movie *King Kong* (1976) contains two characters distinctly identified as a paleontologist and a geologist. Whereas geologists are mostly seen working with rocks directly, geophysicists and paleontologists are typically occupied with seismic measurements and digging up bones.

Gradually, the list of movies has grown. Our initial compilation contained 59 movies and was published in Swedish in the popular magazine of the Geological Society of Sweden in 2013. In 2015, we found 83 movies, closely followed by 92 movies in 2016. In 2018, we thought we had it well covered at 103 movies.

How wrong we were! Around the end of 2020, we got access to parts of the American Film Institute catalogue online, which revealed another 20 older movies. By the end of 2022, More than 10 years after we

started keeping track, we have documented a total of 141 movies, spanning the period 1919–2023 and starring 202 geologists. This article is our fifth compilation, over a decade after our first release. We think we have it well covered now, though naturally we did be happy to be corrected. (In the [supplementary information](#), an Excel file contains the data for movie geologists.)

Geologists in movies through time

The rate of production of geologist movies appears to have a bimodal distribution, with above average 11-year running mean production rates during the periods 1940s–1950s and 1990s–2000s ([Fig. 1](#)).

Geologists feature in movies with various genres ([Fig. 2](#)), following the classification of Internet Movie Database (IMDb). Most movies are assigned multiple genres. The most frequent genres are: adventure (47); drama (40); action (36); and science fiction (35). Only rarely do geologists appear in movies in the genres of news (1), animation (2), musical (2) or history (2). Geologists briefly appeared in romance movies during the 1930s, followed by a peak in western movies that lasted into the late 1950s. At the decline of the western, there was a brief increase in science fiction and horror. Action, adventure and thriller genres all trend upwards to the present, along with a renewed influx in science fiction themed movies. Geologist movies are rarely comedic, and there is a low but steadily increasing amount of drama.

The average demographic of the movie geologists has remained largely unchanged since our initial compilation. Out of 202 geologists, only 22 (11 percent) are female. And of this figure, until 1995, only seven out of 129 (5 percent) cast geologists are female;

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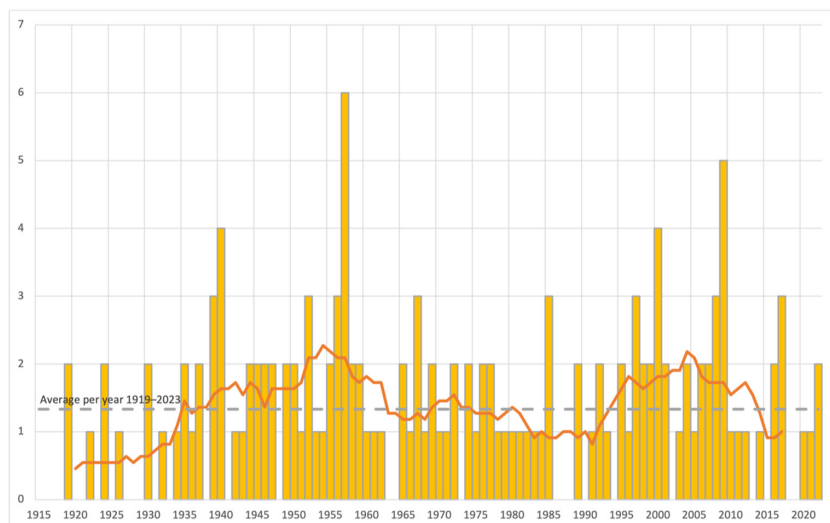


Fig. 1. Bar graph showing the number of geologist movies released per year with the 11-year running mean.

however, since 1996, some 20 percent (15 out of 71) of movie geologists are female.

A total of 172 characters (85 percent) belong to the 'good guys', or at least are not directly evil. Although difficult to determine objectively, we estimate that 43 percent (87 out of 202) of the geologists have roles as major characters in the film. Given the sometimes-overlooked status of our profession, we argue that this is a surprisingly large number. Moreover, even in movies with geologists as minor characters, geology or geological processes are commonly crucial to the plot of the film.



Fig. 2. Bar graphs showing the number of geologist movies released per year within different genres with the 11-year running mean.

In summary, our profession is typically seen as an action-packed adventurous endeavour often perturbed by dramatic events, with minimal indulgence in singing or dancing. While the high count of 'science fiction' could be worrisome for the public perception of our science, a closer analysis reveals that in most of these movies the geologist is at least not guilty of any professional misconduct.

Geologists dying in action

Geologists die at an alarming rate in the movies, often very early on in the movies seen, which requires some deeper analysis. The death percentage varies between 35.7–31.6 percent over our sampling years. Of the 202 geologists in 141 movies, 69 die or are found dead (Fig. 3). Thankfully, this percentage has decreased to 34.2 percent with the latest addition of data. This decrease results from the increased number of movies we found from the 1920s and 1930s. The movie makers during these early days were not so prone to killing the geologist. Only one death (heroic!) in this time slot occurs in *Flying G-Men* (1939).

During the first half of the twentieth century, it was common with movie serials in cinemas, typically as Saturday matinees. The golden age for the serials was between 1936 and 1945. They often had 15 episodes each ending with a 'cliff-hanger', except the final one. Three matinee series feature a geologist: *Flying G-men* (1939), *Valley of Vanishing Men* (1942) and *The Monster and the Ape* (1945). Might we see a recurrence of geologists in serials, but now on streaming platforms?

The manner of geologist movie deaths

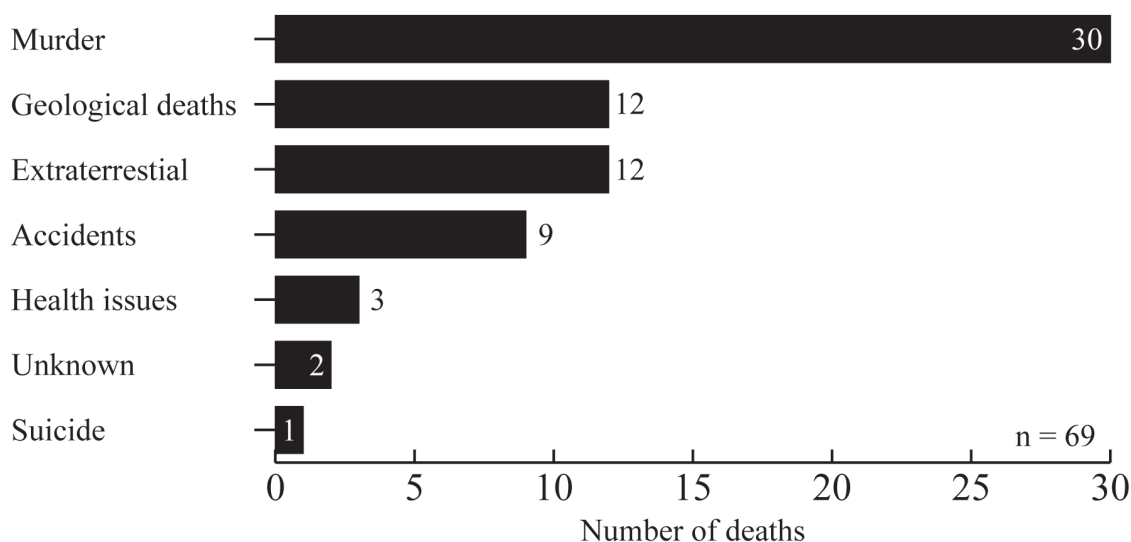
The manners of geologist deaths fall into eight categories (Fig. 3), described below:

Category 1: Murder. A good example is the fatal elimination of 'baddies' by heroes such as James Bond, such as in the movie *Dr No* (1962), where '007' kills the evil petrologist, Prof. Dent.

Category 2: Work related. Typical examples are being hit by a volcanic bomb, falling into a crater or an opening fault, drowning in quicksand or lahar, and incineration in an oil pit. Grim stuff.

Category 3: Killed by or because of Extra-Terrestrial encounters. That is, being killed directly by aliens or assimilation by aliens and consequently killed by former colleagues, such as the geologist in *Prometheus* (2012).

Fig. 3. Bar plot of the causes of death for geologists in movies. Of the 202 geologists, 69 die on the silver screen. The two unknowns are the probable deaths of the geologists in the movie *Gold* (2016) and the movie *Jungle* (2017). Note general accidents and specific traffic accidents fall under the category 'accidents'.



Category 4. Miscellaneous accidents. Such as being crushed by falling crates as in *Deep Star Six* (1989).

Category 5. Traffic accidents. Killed in exotic events like crashing with a spaceship on the asteroid in *Armageddon* (1998).

Category 6. Health issues. Such as heart attack suffered by the geologist in *Equinox* (1970).

Category 7. Suicide. Where it all gets too much, as in *Walkabout* (1971).

Category 8. Unknown. A good example is the fate of the geologist in *Gold* (2016): suicide, or did he fake his death?

The causes of eternal passing are highlighted in the bar plot in Fig. 3. Murder is the most common, with 30 of 69 deaths. The numbers for the second and third reasons are close together, with deaths caused by interaction with extra-terrestrial creatures and unforgiving objects leading to 12 of 69 deaths, and with geological hazards (work related) coming in at 12 of 69 deaths. Accidents, traffic accidents and health issues make a total of 9 of 69 deaths. Notably, we count only one clear case of death by suicide, though in three cases, the fate of the geologist is unknown.

Examples of geologist deaths

Category 1: Murder. Five geologists are murdered in *Money Jungle* (1967). In this movie, five companies compete for an oil concession outside Santa Barbara, California. In the beginning, four geologists die in inventive ways, such as being electrocuted by a golf cart, poisoned when taking medicine, gassed by the fireplace and killed when his toy train explodes. A private eye is hired by the four oil companies that each lost a geologist. Suspicion falls on the fifth company,

particularly when its geologist is murdered just before the private eye was about to question him! The manager of the drill core storage is also murdered but admits before dying that he replaced oil showing core samples with barren ones. The evil plan is to get the concession of the rich block by eliminating geologists that know its potential. It turns out the guilty party is the CEO from another company and the ex-wife of another CEO. This is the worst serial slaughter of geologists in any movie.

Category 7: Suicide. Only one geologist commits suicide in the movies (as far as we know). In *Walkabout* (1971), a father takes his two children to the Australian outback. While the children prepare the picnic and play around, he sits in the car reading the book *Structural geology: An Introduction to Geometrical Techniques* by Donal M. Ragan (1968), which is a well-known text even today. He reads, takes notes, studies geological maps and then suddenly goes nuts—he takes his revolver and starts shooting. He fails to kill his children who flee into the wilderness. Then he sets fire to the car and commits suicide. All this happens in the beginning of the movie and the rest is about the children surviving in the Outback. We hope this does not provoke demands for warning texts on structural geology textbooks!

Evil, fake and deaths

In our survey, 30 of the 202 geologists are evil (criminal) and of these, 23 die (77 percent). This suggests that being evil increases the risk to die, showing that crime does not pay. Perhaps fortunately for the profession, 13 of the evil ones are fake geologists. Actually, there is a total of 19 fake geologists appearing in this survey, and the six good ones are police officers (agents) and treasure hunters. None of these 'good fake' geologists die.

A geologist 'cover' is perceived as a means of providing access to unusual places, doing unusual things. One example is in *The Insider* (1999) when the journalist Lowell calls the FBI from Lincoln in Montana:

Lowell (02:15:15): Just ran into two of your geologists.

Lowell (02:15:18): Geologists whose hands aren't all chewed up.

FBI agent (02:15:21): Lowell, do not... screw this up.

Apparently, despite the inspired disguise, the agents' well-manicured hands blew their cover. Nevertheless, Lowell did not reveal the 'geologists'.

In *Ocean's Thirteen* (2006), Brad Pitt gains access to the no-go underground area of their targeted Las Vegas Hotel, kitted out as a 'geologist', with unkempt hairdo, scraggly beard and thick glasses, clad in a many-pocketed vest and carrying a mysterious instrument. Nobody molests him, because obviously, he must be a geologist!

Heroic geologists

The 85 percent good geologists are doing what geologists do, while also adhering to good moral principles appreciated by society. However, we also find that a good portion of these make an active choice to perform heroic deeds against all odds, like saving loved ones, towns or even the Earth from human or naturally induced calamities. We find 25 movies featuring 38 heroic geologists, or 19 percent of the 202 movie geologists. Most of them survive (28), but sadly, 10 die in action.

The first batch of heroes occur in westerns with themes like oil, cows, railways and spies. The earliest hero appears in the silent movie *Two Women* (1919). The first geologist dying heroically occurs in *Flying G-Men* (1939). Fatally shot in the chest, he asks the 'G-Men' to leave him behind while he, gun in hand, holds up the evil spies. More heroes occur in the monster-laden nuclear bomb-threatening period of 1954–1967. Good examples are *The Black Scorpion*, *The Monolith Monsters* and *The Night the World Exploded*, all from 1957, and each featuring two heroic geologists. The period 1980 to the early 2000s that was dominated by disaster movies boasts *Dante's Peak* (1997) with seven heroic geologists, *Volcano* (1997) with two, and *Armageddon* (1998) and *Deep Core* (2000) both with two. Unfortunately, four of our heroes die in these movies.

Still, the death rate among heroic geologists is only 26 percent as compared to 34.2 percent of all geologists. Obviously, it seems you stand a better chance of survival if you choose to be heroic! However, in *Track of the Moon Beast* (1976), a meteorite fragment from the Moon hits a geologist in the head, whereupon he transforms into a hideous evil lizard with every full moon. After randomly

killing people, he realizes the error of his ways and flees out into the desert to kill himself to protect others. This is of course a self-sacrificing heroic act. But for most of the film, he is decidedly evil, outweighing his heroic deed, so we do not count him among the heroic geologists. He did not have time to commit suicide because he is killed by an arrow with a moon meteorite arrowhead.

Female geologists

The first casting of a female geologist was in the movie *The Night the World Exploded* (1957), but the emergence of female geologists has been slow. More well-known movies are the Bond movie *A View to a Kill* (1983), *Dante's Peak* (1997), *Pitch Black* (2000) and *Wonder Woman* (2020). This slow trickle has now reached a total of 18, and in recent years, the percentage has fallen from 12.4 percent in 2013 to 10.9 percent in 2024. This drop is by the added number of early last century movies totally lacking female geologists. Since 1983, the trend is positive with 21.6 percent female geologists in the movies. Still, there is some way to go if you compare this with the 31.5 percent female members of the Geological Society of London, as an example.

Are all geologists white?

No, but the number of black and minority ethnic geologists is tiny, only six in total. The first black geologist occurs in *Finian's Rainbow* (1968). Clad in a multi-pocketed 'reporters vest' our geologist stuns the white senator by being both black and educated. The only black female geologists we find is in *Bio-Dome* (1996). We find none of Asian, Indian, Pacific or Hispanic descent, whatsoever. This is striking even considering the restriction of our survey to English-spoken US and UK movies. The other movies starring black geologists are *Beware the Blob* (1972), *Evolution* (2001), *2012* (2009) and *Kong: Skull Island* (2017).

Appearances over time

Clothing and general appearance can provide strong signals of what kind of person you are and what you do for a living, and therefore how geologists looked was used to 'state the unspoken' in movies. An obvious example is the 'good guys' wearing big white hats, while 'bad guys' wear black hats in early westerns. One also expects the dress codes of movie geologists to vary with the times and circumstances.

We discern two main 'looks'. One is the archetypal academic professorial type. In older movies, he (as geologists were all portrayed as men at the time) sports a practical sturdy costume suitable for working both outdoors, as well as in not-so-well-heated labs and lecture halls. He may also be naïve and nerdy and look a

bit Einsteinian, as in *Whispering Smith Speaks* (1935) where the elderly professor not only forgets his glasses but even his geological hammer (unforgivable) at an exposure!

The second type is your proverbial field geologist. Oblivious to dirt, oil or weather conditions, they wear workman's clothes like plaid shirts, khaki and heavy-duty boots, helmet, overall or space suit, or whatever clothes are practical and 'unfancy' for the situation. What movie field geologists wear is entirely dependent on where they are, be it ice-cold polar, deserts, wet regions, underground or in space. The clothing indicates a 'whatever gets the job done' attitude, familiar to any real-world geologist. The exception to this is in 'relations' movies like *Another Year* (2010) or *How to Make an American Quilt* (1995), which feature everyday clothes unrelated to geological activity.

Naturally, clothing styles also change over time. In pre-WW2 westerns, field geologists' clothes are often indistinguishable from 'good-guy' Hollywood cowboys, as in *Heritage of the Desert* (1939). Whereas in post-war oil, monster and minerals movies, we see a shift to khaki and one-colour shirts, if costume and tie is not required as in *Attack of the Crab Monsters* (1957). The choice of durable non-fancy clothing shifts over time. But one garment, the multi-pocketed vest, is more persistent over a century of movies than even the plaid shirt!

Geologists are seldom shown as silly, but in *Prospect in Summer Camp* (1979) Chuck Vincent as the prospector wears upturned peak hat on double bandanas, bottle glasses and waving a BIG magnifying glass. But he does save the camp's finances by finding oil under it!

Future geologist movies?

While we have constrained our selection to movies on the silver screen, we expect more geologists to appear in movies and serials produced for streaming platforms such as Netflix, Amazon and Disney.

The portrayal and perceived roles of geologists in movies over 104 years is clearly reflecting changing societal needs, real-world threats and geopolitics over time. We see this from early mineral and oil prospecting, via post-wartime nuclear induced monsters and Cold War era global criminal organizations to recent natural disaster movies. Might the recent shift from disaster movies to fantasy monsters and aliens since 2009 signal a natural disaster fatigue from the real world?

We speculate that the rise of human-induced climate change effects will be fodder for coming movies featuring geoscientists in a wider sense. An indication might be the horror movie *Color out of Space* (2019–20). A meteorite crashes into the garden of a family recently moved to the boondocks, emitting an ethereal weird light. The meteorite's alien influence rapidly corrupts the surroundings, also including the already-from-start weird family. The hydrogeologist

Ward Phillips surveying the area for a hydro dam development happens to be sampling near the impact. He becomes entangled in the family's downward spiral and alien assimilation. Ward hides in the wine cellar from the psychedelic light and the now-raving mad murderous father (Nicholas Cage) while the farmland turns into a final wasteland of ashes through which he shuffles in the end of the movie. Here we see a surviving hydrogeologist, passively hiding (no heroics) from the mess upstairs, not trying to save the day. Afterwards, he emerges into any climate-aware person's nightmare landscape. Although a hydrogeologist does not pass our strict definition for geologist movie, we wonder if we here witness the start of a next generation of movies, catering to a feeling of helplessly watching Mother Nature descend into utter chaos and destruction? We certainly hope not. May our established trend of 85 percent good and often heroic geologists persevere, both in real and movie life!

Acknowledgements

We express our gratitude to all people who have contributed information about movies with geologists.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Data S1.

Suggestions for further reading

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